

# GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER-VII (NEW) EXAMINATION – SUMMER 2024

Subject Code:3170724

Date:15-05-2024

Subject Name:Machine Learning

Time:02:30 PM TO 05:00 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

|     |                                                                                                                                              | Marks |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Q.1 | (a) What is reinforcement learning? State key features of reinforcement learning.                                                            | 03    |
|     | (b) Compare the different types of machine learning.                                                                                         | 04    |
|     | (c) What is IQR? How is it measured? Explain with proper example.                                                                            | 07    |
| Q.2 | (a) How does discretization help us to deal with outliers?                                                                                   | 03    |
|     | (b) Explain qualitative and quantitative data. Differentiate between the two.                                                                | 04    |
|     | (c) What are overfitting and underfitting in context of machine learning models?<br>What is the major cause of overfitting and underfitting? | 07    |
|     | OR                                                                                                                                           |       |
|     | (c) How to improve the performance of a learning model? Explain in details.                                                                  | 07    |
| Q.3 | (a) What is feature engineering?                                                                                                             | 03    |
|     | (b) Define sampling with replacement. Give proper example.                                                                                   | 04    |
|     | (c) Explain naïve bayes classifier with an example.                                                                                          | 07    |
|     | OR                                                                                                                                           |       |
| Q.3 | (a) When can a feature be termed as irrelevant? How can it be measured?                                                                      | 03    |
|     | (b) What is sampling without replacement? Give proper example.                                                                               | 04    |
|     | (c) Explain Bayesian belief network.                                                                                                         | 07    |
| Q.4 | (a) Explain the classification learning steps.                                                                                               | 03    |
|     | (b) What are the factors determining the effectiveness of SVM?                                                                               | 04    |
|     | (c) Discuss the decision tree algorithm in detail.                                                                                           | 07    |
|     | OR                                                                                                                                           |       |
| Q.4 | (a) Why the kNN algorithm is called a lazy learner?                                                                                          | 03    |
|     | (b) Describe the strengths and weaknesses of decision tree.                                                                                  | 04    |
|     | (c) Explain the difference between linear and logistics regression with example.                                                             | 07    |
| Q.5 | (a) Discuss the strengths and weaknesses of the k-means algorithm.                                                                           | 03    |
|     | (b) Explain hierarchical clustering.                                                                                                         | 04    |
|     | (c) Explain the different types of activation functions used in neural network.                                                              | 07    |
|     | OR                                                                                                                                           |       |
| Q.5 | (a) Differentiate between supervised learning versus unsupervised learning.                                                                  | 03    |
|     | (b) Explain density-based methods.                                                                                                           | 04    |
|     | (c) Explain the backpropagation algorithm. What are the limitations of this algorithm?                                                       | 07    |